# Experiment 3: Stepper Motor using AT89C51 using Proteus

**Aim:**

To control a stepper motor using AT89C51 microcontroller in Proteus.

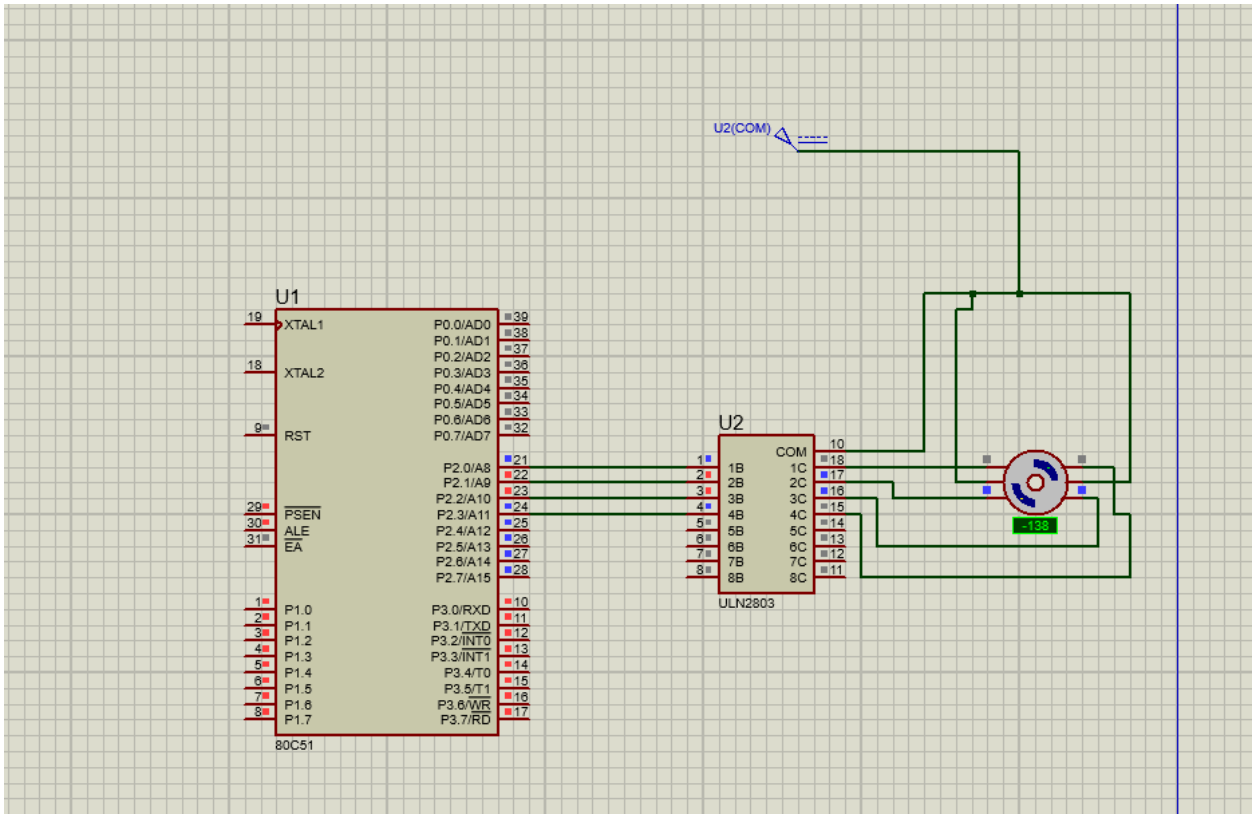
**Procedure:**

1. Open Proteus and create a new circuit.
2. Place AT89C51, stepper motor and driver circuit.
3. Connect the driver to the microcontroller port.
4. Write and compile the stepper motor control program.
5. Load the HEX file into AT89C51.
6. Run the simulation and observe the motor rotation.

**ANTI CLOCKWISE**

main.asm

```
1   ORG 0000H
2
3   UP:       MOV P2,#03H
4             ACALL DELAY
5
6             MOV P2,#06H
7             ACALL DELAY
8
9             MOV P2,#0CH
10            ACALL DELAY
11
12            MOV P2,#09H
13            ACALL DELAY
14
15            SJMP UP
16
17  DELAY:    MOV R4,#18
18
19  H1:       MOV R3,#255
20  H2:       DJNZ R3,H2
21            DJNZ R4,H1
22
23            RET
24
25  END
```




**CLOCK WISE**

```
ORG 0000H

UP:         MOV P2, #09H
            ACALL DELAY

            MOV P2, #0CH
            ACALL DELAY

            MOV P2, #06H
            ACALL DELAY

            MOV P2, #03H
            ACALL DELAY

            SJMP UP

DELAY:      MOV R4, #18

H1:         MOV R3, #255

H2:         DJNZ R3, H2
            DJNZ R4, H1

            RET

END
```

**Result:**

The stepper motor was successfully controlled and rotated using AT89C51.